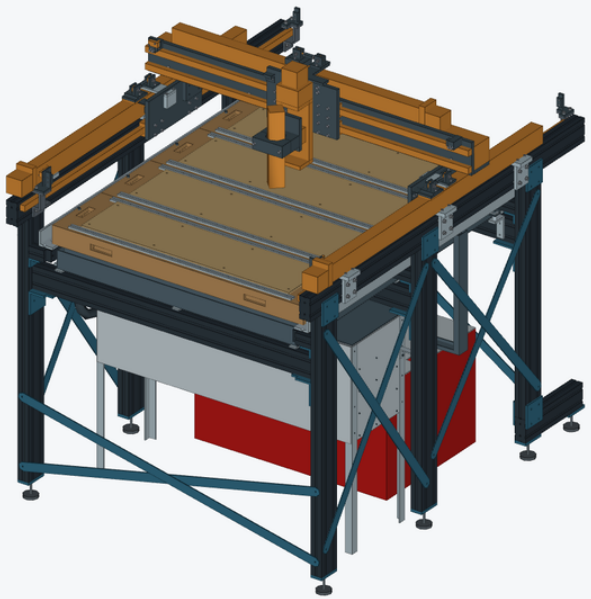
4080 frame. Removable dry bed. Steel plasma pan.

Small machine: 800 X / 1,000 Y / 100 Z nominal travel. Selective billet mounts retained.

XR1 | 4080 WOOD + PLASMA TABLE

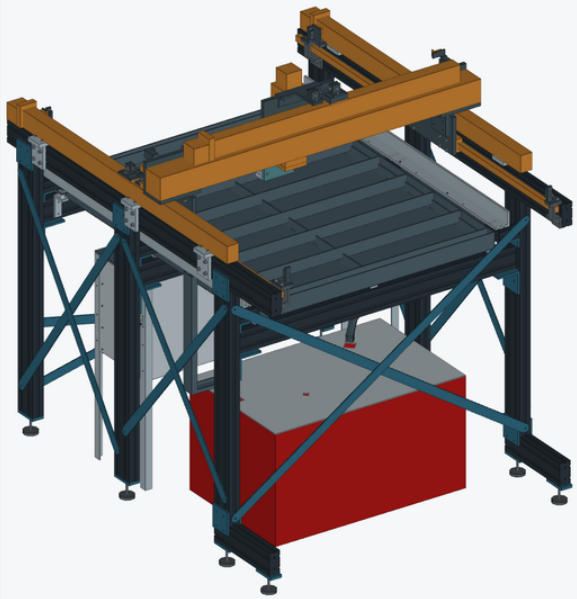
800 X / 1,000 Y / 100 Z nominal travel • Engineering revision

01 WOODWORKING



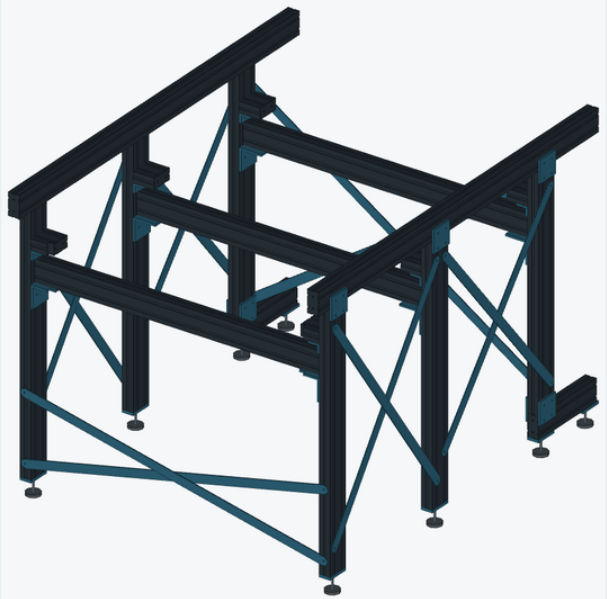
Removable dry bed with aluminum T-slot workholding.

02 PLASMA



Steel hot zone; dry bed in covered storage.

03 FRAME



Six continuous posts; no second full-length frame tier.

Geometry, loads and conversion paths are checked separately.

Supplier interfaces and physical acceptance tests remain release gates. No prices in this revision.

Actual CAD solids shown. Tool, torch and cable envelopes remain subject to measurement.

Current design replaces the earlier cart and frame proposals. No new pricing or shopping.

Geometry and conditional structural targets

The pan supports water and steel stock. Separate ledgers support woodworking loads.

Frame

12.3574 m finished 4080; 19 cuts

Supports

Six posts; short rear bases; eight main feet

Dry carriers

Four 932 x 257 x 70.8 cassettes

Wear field

770 x 960 enclosing field; track/seam keep-outs

Construction

12.7 ply skins; 39.05 core; 6.35 MDF wear

Hot pan

894 x 1096 outside; 1.9 nominal plain steel

Slats

Eight 880 x 40 x 1.9; replaceable

Owned stock

Two 304.8-square Al blanks; 75 nested blanks

WATER VOLUMES

Operating level	**37.05 L**
Overflow begins	**45.68 L**
Full to brim	**56.23 L**
Estimated gravity-drain residue	**0.45 L**

Selected red 20-gallon tote: allow 712 x 470 x 389 externally. Require 65 L measured usable below the chosen maximum fill mark. Actual liquid load and temperature rating remain unverified.

100 N VERTICAL SCREEN

Selected plywood case: approximately 0.230 mm ideal bed + ledger + upper-beam subtotal. The framework/bed target is 0.30 mm, leaving about 0.070 mm for omitted joint/contact and other compliance.

Weaker plywood sensitivity is about 0.311 mm and exceeds that target. This is a component calculation, not assembled-frame FEA or demonstrated tool-to-work accuracy.

Extrusion section, grain direction, glue quality and joint preload are conditions of these results.

Audit evidence and release gates

A successful CAD check covers its stated scope; it does not qualify the complete machine.

DIGITAL REVIEW PASSED WITHIN STATED SCOPES

STEP round trips: frame 89 solids; wood 811; plasma 890.

Display-pose external intersections and frame/bed self checks are recorded separately.

XYZ external travel uses full translation bounds and exact ledger-envelope proofs; internal relative motion is sampled.

Bed conversion checks 32 complete-path segment enclosures within the machine plan.

Owned-aluminum and profile-stock nests include cut gaps; deep billet gantry brackets and clevises are not assumed owned.

REQUIRED BEFORE FABRICATION / CAM RELEASE

Verify delivered module mounts, guide geometry, lead, Z retention, actual tool/torch projection and the cutter interface.

Finish mounted heat shields, reservoir hatch/removal access and flat-pattern/half-lap machining details before manufacturing.

Qualify extrusion, fasteners, assembled stiffness and stability; cold-test water flow/capacity and thermally test protected storage.

Integrate the requested retractable ATC using the actual kit and Z stack. It is not installed or qualified in these assemblies.

USD 1,000 remains the complete-table budget constraint. It is not a verified delivered quote.